

first name

last name

email

In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. Find $\lim_{x \rightarrow 6} \frac{8}{|x-6|}$

$\frac{8}{0}$

$\lim_{x \rightarrow 6} |6-6|$

$\lim_{x \rightarrow 6} 8 = 8$

$\lim_{x \rightarrow 6} |x-6| = |6-6| = |0| = 0$

x	y
5.9	$\frac{8}{ 5.9-6 } = \frac{8}{ -0.1 } = \frac{8}{0.1} = 80$
5.99	800

x	y
6.1	$\frac{8}{ 6.1-6 } = \frac{8}{ 0.1 } = \frac{8}{0.1} = 80$
6.01	800

$\lim_{x \rightarrow 6^-} \frac{8}{|x-6|} = \infty$

$\lim_{x \rightarrow 6^+} \frac{8}{|x-6|} = \infty$

$\lim_{x \rightarrow 6} \frac{8}{|x-6|} = \infty$

If you'd prefer, write $f(x)$ after limit instead of $\frac{f}{|x-6|}$ after limit

Cannot split before canceling.

(10pts) 2. Find $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$

$\frac{0}{0}$

$\lim_{x \rightarrow 2} (2^2 + 2 - 6)$

$\lim_{x \rightarrow 2} (2 - 2)$

$\lim_{x \rightarrow 2} (x^2 + x - 6) = 2^2 + 2 - 6 = 0$

$\lim_{x \rightarrow 2} (x - 2) = 2 - 2 = 0$

$= \lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{x-2}$

$= \lim_{x \rightarrow 2} (x+3)$

$= 2+3$

$= 5$

$\lim_{x \rightarrow 2} (2+3)$

(10pts) 3. Determine by definition if

$f(x) = \begin{cases} 2x + 5 & \text{if } x \leq 1 \\ 6 - 3x & \text{if } x > 1 \end{cases}$

is continuous at $x = 1$.

$f(1) = 2(1) + 5 = 7$

$f(1)$ exists.

$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2x + 5) = 2(1) + 5 = 7$

$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (6 - 3x) = 6 - 3(1) = 3$

$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$ so $\lim_{x \rightarrow 1} f(x)$ dne so f is not continuous at $x = 1$.